

Kunal Maheshwari

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SUMMARY

OPT-EAD holder with full-time work authorization and over 4 years of experience in Software Development and Data Engineering, holding a Master of Science in Computer Science. Proven ability to design and implement scalable microservices and distributed data pipelines using Java, Python, Spark, SQL, AWS, and Azure, with expertise in CI/CD automation, J2EE, and advanced data platforms. Adept at optimizing backend systems, driving performance improvements, and contributing to privacy-preserving AI applications, aligning with a Java Backend Architect role.

EDUCATION

Central Michigan University | Mount Pleasant, MI Aug 2024 - Dec 2025
Master of Science | Computer Science

Lovely Professional University | Punjab, India Aug 2016 - May 2020
Bachelor of Technology | Computer Science and Engineering

EXPERIENCE

Software Developer | NTT DATA | Bengaluru, India (Remote) August 2022 - August 2024

- Engineered high-performance RESTful APIs and SOAP microservices using Java 8/11, Spring Boot, and Azure Functions, achieving 99.99 percent uptime and managing over 50,000 daily API requests.
- Automated CI/CD pipelines via Jenkins, Maven, and Azure DevOps for seamless deployment across Azure Cloud and Oracle WebLogic environments.
- Modernized legacy systems by migrating to Log4j 2.x and enhancing monitoring, reducing issue resolution time by 20 percent through improved error handling.
- Led a critical EDX platform upgrade, streamlining application architecture and dependency management to achieve 50 percent faster delivery cycles.
- Collaborated in Agile and Scrum teams to deliver FHIR-compliant healthcare solutions, leveraging Git, SVN, and JUnit testing for robust backend development.

Senior Analyst | Capgemini | Mumbai, India (Remote) October 2021 - August 2022

- Developed robust Java and EJB-based enterprise applications and RESTful web services for automotive clients, focusing on backend development.
- Enhanced system performance by 90 percent through strategic SQL query optimization, database tuning, and modernization of legacy Java code.
- Implemented CI/CD pipelines utilizing Jenkins and GitLab, incorporating automated testing and quality gates for improved code reliability.
- Mentored junior developers by conducting thorough code reviews, providing debugging assistance, and promoting clean coding practices.

Analyst | Capgemini | Mumbai, India (Remote) July 2020 - October 2021

- Maintained high-availability enterprise systems with over 99 percent uptime through proactive monitoring and performance optimization, including APM tools.
- Improved SQL data pipelines and database operations, achieving a 10 percent enhancement in query performance and strengthening data integrity.
- Implemented comprehensive unit testing using JUnit4, reaching 99 percent code coverage for critical backend components.
- Supported release planning and sprint execution using JIRA within Agile development teams, contributing to efficient logistics.

PROJECTS

NYU Taxi Data Analysis Platform January 2024 - May 2025

- Built a distributed Spark-based data pipeline to process and analyze over 8 GB of NYC taxi trip data in near real time, showcasing ETL pipeline expertise.
- Developed an interactive Flask web application for data exploration with dynamic filtering and visualization, demonstrating backend development capabilities.
- Containerized ingestion, processing, and analytics components using Docker for reproducible deployments.
- Optimized Spark memory usage and partitioning strategies, reducing overall job runtime by 40 percent.
- Automated cloud infrastructure provisioning using Terraform across AWS services including S3, RDS, ECS, Lambda, and QuickSight for infrastructure automation.

Privacy-Preserving AI Email Client (Capstone Project) January 2025 - December 2025

- Engineered a fine-tuning pipeline for T5-based transformer models to perform email summarization and classification, integrating AI and backend development.
- Applied parameter-efficient fine-tuning using LoRA to support low-resource training under privacy constraints.
- Implemented model evaluation using ROUGE metrics and integrated quantization for secure local deployment, contributing to privacy-preserving AI applications.

SKILLS

- Programming Languages: Java, Python, SQL, Kotlin, PL/Sql.
- Frameworks and Libraries: Spring Boot, Spring Framework, J2EE, Apache Spark, PySpark, Hugging Face, T5, LoRA, JUnit, JMS, Jpa.
- Web Services: RESTful API, SOAP API, GraphQL.

- Cloud Platforms: AWS (EC2, RDS, S3, Lambda), Azure (Functions, DevOps, ServiceBus), Google Cloud Platform.
- Databases: PostgreSQL, Oracle Database, IBM DB2, RDS.
- Tools and Technologies: Docker, Kubernetes, Jenkins, Github Actions, Git, SVN, Maven, JIRA, Confluence, Tableau, DBeaver, Terraform, Doors.
- Concepts: ETL Pipelines, Data Warehousing, Data Analytics, Agile, Scrum, Test Driven Development, Object Oriented Programming, CI/CD, Microservices, Large-scale data platforms, Infrastructure automation, Privacy-preserving LLM-based applications, Caching, Backend Development.
- Data Analysis and Visualization: Tableau, QuickSight.